

AYUSH RAJ TIWARY

AI/ML Engineer | Full-Stack Developer | LLM Applications
ayushrajtiwary07@gmail.com | +91 7848956921 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Developer with 3 deployed production projects. Built NEURO-CORE — 3 PyTorch models (CNN, ResNet-18, EfficientNet-B0 at 95.75%) for MRI tumor classification with Grad-CAM and in-browser MediaPipe gesture control. Built Spice Garden, a production AI dining platform with 8 LangChain.js agents, Cohere RAG over pgvector, real-time Socket.io ordering, and Razorpay payments. Experienced with Docker, FastAPI, Next.js, Redis, and Prisma. SIH 2025 participant and Google Gen AI Exchange 2025 developer.

TECHNICAL SKILLS

AI and ML: PyTorch, CNN, ResNet-18, EfficientNet-B0, Grad-CAM, Focal Loss, Transfer Learning, TensorFlow, OpenCV, MediaPipe, NumPy, Pandas, Scikit-learn

LLM and GenAI: LangChain.js, Groq API, Cohere Embeddings, pgvector, RAG Pipelines, Prompt Engineering, Multilingual NLP

Frontend: React.js, Next.js, Three.js, Socket.io Client, Zustand, Tailwind CSS, REST APIs, PWA

Backend: Node.js, Express.js, FastAPI, PostgreSQL, Prisma ORM, MongoDB, Redis, WebSocket

DevOps and Cloud: Docker, Supabase, Vercel, Render, Netlify, Cloudinary, Git, GitHub, Postman

Languages: Python, JavaScript, Java, C, C++

Core CS: Data Structures and Algorithms, OOP, System Design, Model Deployment

PROJECTS

NEURO-CORE — Brain Tumor MRI Detection with Gesture Control

[GitHub](#) | [Live Demo](#)

Python · PyTorch · CNN · ResNet-18 · EfficientNet-B0 · Grad-CAM · Focal Loss · FastAPI · MediaPipe · Three.js · Groq · Render

- Trained 3 independent PyTorch models on 7,200 brain MRI scans using Focal Loss: Custom CNN (89.94%), ResNet-18 (95.12%), and EfficientNet-B0 (95.75% best — only 5.3M params vs 11.3M ResNet-18).
- Implemented per-model Grad-CAM heatmaps (each model uses its validated target layer) predictions.
- Built in-browser MediaPipe Hands gesture engine (WebAssembly, zero server calls) — pinch-to-click, dwell-select, two-hand scroll, pinch-zoom, air-grab toggle — enabling fully sterile, contactless diagnostic review; served via FastAPI on Render.

Spice Garden — AI-Driven Smart Dining Assistant

[GitHub](#) | [Live Demo](#)

Next.js · Node.js · PostgreSQL · pgvector · Cohere · LangChain.js · Groq · Redis · Socket.io · Prisma · Razorpay · Cloudinary · Docker · Vercel · Render

- Architected 8-agent LangChain.js system (Orchestrator, Recommendation, Upsell, Sentiment, Memory, Multilingual NLU, Group Coordinator, Order Validation) routing intents to specialized agents.
- Built real RAG pipeline using Cohere embeddings (384-dim) in pgvector on Supabase; supports multilingual NLU (English, Hinglish, Telugu), dietary profile filtering, allergy alerts, and cross-session memory.
- Delivered 25+ features: Socket.io real-time group ordering, Razorpay payments, WhatsApp notifications, PWA, revenue forecasting, table reservations, inventory management, and multi-restaurant support; 25-model Prisma schema; containerized with Docker.

Air Gesture Recognition System

[GitHub](#)

React.js · Node.js · Express.js · OpenCV · REST API

- Implemented real-time touchless image transfer across browser tabs using a Grab and Drop gesture interaction model powered by OpenCV hand tracking.
- Built low-latency Node.js and Express REST API backend with in-memory caching, reducing manual drag-and-drop effort by approximately 60%.

HACKATHONS AND ACHIEVEMENTS

Smart India Hackathon 2025 — Suraksha Setu

Full-Stack Developer

- Built MERN stack tourist safety platform with real-time location-based alerts, emergency notifications, and Google Maps API integration — competed at college level.

Google Gen AI Exchange 2025 — Mental Health Assistant

AI Developer

- Designed and deployed an AI-powered conversational mental health assistant using Gen AI APIs for natural language understanding and response generation within hackathon constraints.

EDUCATION

B.Tech in Computer Science Engineering

2023 – 2027

Jharkhand Rai University, Ranchi | Specialization: AI and ML | CGPA: 7.3 (up to 5th sem) | Current: 6th sem